

Ctrl + Incision - Sprint 1: Retrospective

Sprint Duration: 2.5 weeks

Team Size: 5

Goal: Deliver a functional vertical slice with tracking simulation, UI workflow, modularized backend, and database foundation.

1. What Went Well

✓ **Team Collaboration & Communication**

- Everyone contributed meaningfully.
- Meetings were productive; PRs were merged efficiently.
- Good division of responsibilities:

UI workflow (Muneeb)

FakeTracker + tracking pipeline (Delaney)

Database system (Hannah)

Refactoring + cleaned file structure (Elise)

Refactoring + modularization (Janki)

Architecture & integration (team-wide)

✓ **Functional Vertical Slice Delivered**

The app now supports:

- Synthetic 6-DOF tracking
- Displaying pose data in UI
- Calibration workflow structure
- Camera and overlay modules
- SQLite database logging

✓ **UI Workflow Strong Foundation**

- Clear step-by-step calibration workflow
- Status indicators
- Scroll fixes
- Logical grouping of intrinsic, pivot, and hand-eye steps

✓ **Refactoring Made Code Maintainable**

- OverlayApp split into two modules
- Cleaner interfaces
- Easier to extend in Sprint 2

✓ **Advisor Positive Feedback**

- Very satisfied with architecture
- Acknowledged strong participation
- Recognized usefulness of the platform

2. What Didn't Go Well

Mathematical Concepts Not Yet Deeply Understood

Team needs deeper understanding of:

- Transformation chains
- Matrix multiplication order
- Hand-eye & probe calibration theory
- Error propagation
- ❖ These will become mandatory in Sprint 2.

No Hardware Testing Yet

- FakeTracker masks issues that real devices may show.
- Latency was good, but video/tracking synchronization untested.

Limited Exploration of Vision-Based Tip Detection

- While suggested by advisor, not implemented yet:
- RGB vs HSV models
- Hough transform
- scikit-image detection
- ML-based marker tracking

Manual Testing & Limited Automation

- All tests are currently manual demos.

3. Team Learnings

- UI and backend should be finalized early to avoid re-wiring.
- Transformation math is foundational; cannot be postponed.

- Modular design dramatically reduces refactors.
- Early advisor feedback prevents misaligned development.
- Interfacing hardware requires planning well ahead.

4. Improvements for Sprint 2

A. Begin 3D Ultrasound Reconstruction Early

This is the hardest sprint:

- Volume building
 - Frame-to-probe transformations
 - Performance optimization
- Synchronization with tracking stream

B. Implement & Test Real Calibration Algorithms

- Intrinsic calibration
- Pivot calibration
- Probe calibration
- Hand-eye calibration
- Verify transformations visually

C. Integrate Real Tracking Hardware

- NDI Polaris/Aurora
- ROM file loading
- Latency + jitter handling

D. Strengthen Vision-Based Tip Tracking (Optional / Research-Oriented)

- Detect circular stylus tip
- Hough transform
- ML-based detection
- Test with pen + marker at home

E. Improve Logging & UI Feedback

- Timestamped logs
- Session-based history
- Popup errors
- Progress indicators for long calibrations

F. Expand Documentation

- Transformation chain diagrams
- Probe→Image mapping
- 3D reconstruction algorithm design

5. Sprint 2 Goals

- Start implementing 3D US reconstruction module
- Integrate real tracking hardware
- Finish pivot, intrinsic, hand-eye, and probe calibration pipelines
- Establish transformation graph for full system
- Weekly advisor meetings for feedback
- Gantt chart until April 1 with milestones
- Prepare for early March/April lab presentations

Conclusion

Sprint 1 delivered a strong foundation — modular, stable, and functional. Sprint 2 will focus on the core scientific and engineering challenge of the entire project: 3D ultrasound reconstruction and calibration accuracy.

The team has the structure and momentum to succeed, provided we maintain clear planning, deepen mathematical understanding, and begin real hardware testing early.